HIROFUMI INAGUMA

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RESEARCH INTERESTS

Automatic speech recognition (ASR)
- End-to-end speech recognition
- Multilingual end-to-end speech recognition
- Language modeling
- Transfer learning, semi-supervised training
- Online/Streaming ASR

Speech translation
- End-to-end speech translation
- Multilingual end-to-end speech translation

EDUCATION

Ph.D. in Computer Science, Kyoto University, Kyoto, Japan *April 2018 - Present*
Department of Intelligence Science and Technology, Graduate School of Informatics
- Supervisor: Prof. Tatsuya Kawahara

M.E. in Computer Science, Kyoto University, Kyoto, Japan *April 2016 - March 2018*
Department of Intelligence Science and Technology, Graduate School of Informatics
- Thesis title: Joint Social Signal Detection and Automatic Speech Recognition based on End-to-End Modeling and Multi-task Learning
- Supervisor: Prof. Tatsuya Kawahara

B.E. in Computer Science, Kyoto University, Kyoto, Japan *April 2012 - March 2016*
- Supervisor: Prof. Tatsuya Kawahara

WORK EXPERIENCES

Microsoft Research, Redmond, WA, USA *July 2019 - October 2019*
Research Internship (3 months)
- Mentor: Yifan Gong, Jinyu Li, Yashesh, Gaur, and Liang Lu

Johns Hopkins University, Baltimore, MD, USA *July 2018 - September 2018*
Visiting student (2.5 months)
- Worked on end-to-end speech recognition and translation
- Participated in the JSALT workshop (topic: multilingual end-to-end speech recognition)
- Participated in the IWSLT2018 evaluation campaign on the end-to-end speech translation track
- Mentor: Prof. Shinji Watanabe

IBM research AI, Tokyo, Japan *September 2017 - November 2017*
Research Internship (2 months)
- Worked on end-to-end ASR systems
- Mentor: Gakuto Kurata, and Takashi Fukuda

Recruit Co.,Ltd., Tokyo, Japan *February 2017*
Research Intern (2 weeks)
- Worked on data analysis and image classification competition
- Won the 1st place

CyberAgent, Inc., Tokyo, Japan *September 2016 - October 2016*
Research Intern (1 months)
- Worked on the development of music recommendation systems

TECHNICAL STRENGTHS

Programming language	Python, Bash, C/C++, Java, LaTeX, PHP, Javascript, CSS, HTML
Software & Tools	ESPnet, Kaldi, Docker
Deep learning frameworks	Pytorch, Tensorflow, Chainer

LANGUAGE SKILL

Japanese (native), English (fluent)

AWARDS

Microsoft Research Asia Ph.D. Fellowship Award, from Microsoft Research Asia (MSRA), October 2019

Yamashita SIG Research Award, from Information Processing Society of Japan (IPSJ), March 2019
- Paper title: "An End-to-End Approach to Joint Social Signal Detection and Automatic Speech Recognition"

Yahoo! JAPAN award (best student paper), from SIG-SLP, June 2018

Research Fellowship for Young Scientists (DC1), from Japan Society for the Promotion of Science (JSPS), April 2018 - March 2021

Student award, from the Acoustical Society of Japan (ASJ), March 2018

Student award, from the 79th of National Convention of Information Processing Society of Japan (IPSJ), March 2017

ACADEMIC SERVICES

Reviewer: ICASSP2020, ICASSP2019

RESEARCH PROJECTS

Multilingual end-to-end speech translation
Proposed an effective framework for the multilingual end-to-end speech translation based on a single sequence-to-sequence model, and showed significant improvements of the translation performances compared to the conventional bilingual models.

Transfer learning of multilingual end-to-end ASR with language model fusion
Developed the language-independent end-to-end ASR system based on a single sequence-to-sequence model, and successfully adapted to unseen languages with the external language model.

Acoustic-to-ward sequence-to-sequence ASR without out-of-vocabulary (OOV) words
Proposed a sequence-to-sequence ASR framework which directly outputs words, and resolved the data sparseness and unknown words (UNK) problems by multi-task learning with the character-level ASR task. Character-level hypotheses are also leveraged during the inference stage to recover OOV words.

Joint social signal detection and ASR
Incorporated social signal detection (e.g., laughter, filler, back-channels, and disfluencies) into the end-to-end ASR task, and proposed a unified framework for jointly modeling both tasks.

PUBLICATIONS (REVIEW PAPER, FIRST AUTHOR)

Hirofumi Inaguma, Yashesh, Gaur, Liang Lu, Jinyu Li, and Yifan Gong, "Minimum latency training strategies for streaming sequence-to-sequence ASR", IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2020 (Acceptance Rate: 47%, **selected for oral session**).

Hirofumi Inaguma, Kevin Duh, Tatsuya Kawahara, and Shinji Watanabe, "Multilingual End-To-End Speech Translation", IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), 2019. (Acceptance Rate: 144/299=48.1%)

Hirofumi Inaguma, Jaejin Cho, Murali Karthick Baskar, Tatsuya Kawahara, and Shinji Watanabe, "Transfer Learning of Language-Independent End-to-End ASR with Language Model Fusion", IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019. (Acceptance Rate: 1774/3815=46.5%)

Hirofumi Inaguma, Masato Mimura, Shinsuke Sakai, and Tatsuya Kawahara, "Improving OOV Detection and Resolution with External Language Models in Acoustic-to-Word ASR", IEEE Spoken Language Technology Workshop (SLT), 2018. (Acceptance Rate: 150/257=58.3%)

Hirofumi Inaguma, Xuan Zhang, Zhiqi Wang, Adithya Renduchintala, Shinji Watanabe, and Kevin Duh, "The JHU/KyotoU Speech Translation System for IWSLT 2018", 15th International Conference on Spoken Language Translation (IWSLT), 2018.

Hirofumi Inaguma, Masato Mimura, Koji Inoue, Kazuyoshi Yoshii, and Tatsuya Kawahara, "An End-to-End Approach to Joint Social Signal Detection and Automatic Speech Recognition", International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2018. (Acceptance Rate: 1406/2830=49.7%)

Hirofumi Inaguma, Koji Inoue, Masato Mimura, and Tatsuya Kawahara, "Social Signal Detection in Spontaneous Dialogue Using Bidirectional LSTM-CTC", 18th Annual Conference of International Speech Communication Association (Interspeech), 2017. (Acceptance Rate: 799/1582=52.0%)

Hirofumi Inaguma, Koji Inoue, Shizuka Nakamura, Katsuya Takanashi, and Tatsuya Kawahara, "Prediction of Ice-breaking between participants using prosodic features in the first meeting dialogue", International Conference Multimodal Interaction workshop on Advancements in Social Signal Processing for Multimodal Interaction (ASSP4MI), 2016.

PUBLICATIONS (REVIEW PAPER, CO-AUTHOR)

Shigeki Karita, Nanxin Chen, Tomoki Hayashi, Takaaki Hori, **Hirofumi Inaguma**, Ziyan Jiang, Masao Someki, Nelson Enrique Yalta Soplin, Ryuichi Yamamoto, Xiaofei Wang, Shnji Watanabe, Takenori Yoshimura, and Wangyou Zhang, "A Comparative Study on Transformer vs RNN in Speech Applications'", IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), 2019.

Jaejin Cho, Shinji Watanabe, Takaaki Hori, Murali Karthick Baskar, **Hirofumi Inaguma**, Jesus Villalba, Najim Dehak, "Language Model Integration Based on Memory Control for Sequence to Sequence Speech Recognition", IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019.

Masato Mimura, Sei Ueno, **Hirofumi Inaguma**, Shinsuke Sakai, and Tatsuya Kawahara, "Leveraging Sequence-to-Sequence Speech Synthesis for Enhancing Acoustic-to-Word Speech Recognition", IEEE Spoken Language Technology Workshop (SLT), 2018.

Sei Ueno, **Hirofumi Inaguma**, Masato Mimura, and Tatsuya Kawahara, "Acoustic-to-Word Attention-Based Model Complemented with Character-level CTC-Based Model", IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2018.

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